

- X a) The heat loads in an air conditioned room are 30,000 k cal/hr. sensible heat and 20,000 K cal/hr. latent heat. Condition of air leaving the room should not exceed 27°C DBT and 60% RH. Find condition of air entering the room if quantity of air flow at the exit at the room is 255 m³/minute.
- b) Draw a neat diagram of air conditioning system required in winter season. Explain different components in circuit.

BTS 129(J)

*B.Tech. Degree IV Semester (Supplementary) Examination in
Mechanical Engineering (CAD/CAM), June 2001*

ME 402 THERMAL ENGINEERING I
(Before 1998 admissions)

Time: 3 Hours

Max. Marks: 100

*(Thermodynamic property tables and charts are permitted.
All questions carry equal marks)*

- I a) What do you understand by solid injection ? What are its advantages over air injection system ?
- b) Briefly explain i) burning time loss factor ii) Heat loss factor iii) Exhaust blow down factor.
- c) Explain the method of conducting a morse test.
- d) Which are the factors influencing volumetric efficiency of IC engines?
- OR**
- II A single cylinder 4 stroke cycle oil engine works on diesel cycle. Following readings were taken when engine was running at full load. Area of indicator = 3cm², Length of diagram = 4 cm Spring constant = 10 bar/cm²-cm. Speed of the engine = 400 rpm, Load on the brake = 380 N Spring reading = 50 N, Diameter of brake drum = 120 cm, Fuel consumption = 2.8 kg/hr. CV of fuel = 42,000 KJ/Kg. Dia of cylinder = 16 cm, Stroke of piston = 20 cm. Find (i) FP of engine (ii) Brake thermal efficiency (iii) Mechanical efficiency d/BMEP.
- III a) Explain off - design flow characteristics of convergent - divergent nozzle.
- b) Explain effect of friction in nozzle flow with T-S & h - s diagram.

(P.T.O)

- c) Steam at 30 bar and specific volume $0.0993 \text{ m}^3/\text{kg}$. enters a convergent - divergent nozzle with negligible speed and expands into a space at 4 bar. Calculate the throat and exit areas for a mass flow rate of 1 kg/s. Assume steam expands isentropically according to the law $PV^{1.3} = \text{constant}$.

OR

- IV a) Explain principle of working of an impulse turbine and reaction turbine with diagram.
- b) 10 Kg. of steam enters an impulse turbine stage. The Jet velocity of steam through nozzles is 800 m/s. Blade velocity is 200 m/s. Nozzle angles is 60° and blade angle at out let is 25° . Relative velocity coefficient is 0.8. Calculate (i) blading efficiency (ii) power developed (iii) Frictional energy of blades.
- V A single acting, single stage reciprocating air compressor, 250mm bore 350mm stroke runs at 3 rps. The suction and delivery pressures are 0.98 bar and 6 bar respectively. Calculate the theoretical power required to run it under each of the following conditions of compression.
(i) Isothermal (ii) polytropic $n = 1.3$, (iii) adiabatic $\gamma = 1.4$
Neglect clearance. Also calculate isothermal efficiency for (ii) and (iii).

OR

- VI a) Describe working of an axial flow compressor with the help of schematic.
- b) A centrifugal compressor work with total head pressure ratio of 3.5 : 1. The speed of the compressor is 18000 rpm. Inlet stagnation temperature is 15°C and slip factor = 0.91., power input factor = 1.04 and Isentropic efficiency = 0.85. Calculate the outer diameter of the impeller.
- c) Explain about surging in axial flow compressors.

- VII a) Draw a neat diagram of vapour compression refrigeration system and explain its working.

(P.T.O)

- b) An air refrigerator working on Bell Coleman cycle takes air from cold chamber at 1 bar and -5°C and compresses to 6 bar following the law $PV^{1.25} = C$. Compressed air is cooled to 37°C in the cooler before entering into expander. Expansion is isentropic. Determine (i) COP of cycle (ii) Mass of air circulated per minute if 500 kg. of ice is produced per day at 0°C when water is supplied at 20°C and (iii) Refrigeration capacity of plant in tons, Neglect clearances in compressor and expander $\gamma = 1.4$, $C_p = 1 \text{ KJ/Kg. of air}$. Latent heat of Ice = 335 KJ/Kg. $C_p (\text{water}) = 4.18 \text{ KJ/Kg}$.

OR

- VIII a) Draw a sketch of ammonia refrigeration system and explain its working.
- b) Capacity of a refrigerator is 600 tons when working between -5°C and 20°C . Find the weight of Ice produced within 24 hours when water is supplied at 10°C . Find minimum HP required.
- c) Which are the common refrigerants used. Compare their properties.
- IX a) Define comfort and explain how comfort chart is constructed. Show best summer and winter design conditions zones in comfort chart.
- b) Ten grams of moisture per kg. of dry air is removed from atmospheric air when it is passed through an air conditioning system and its temperature becomes 20°C . Atmospheric conditions are 40°C DBT and 60% RH. Find following (i) Relative humidity (ii) Wet bulb temperature (iii) Dew point temp.
Take air pressure = 1.033 kg/cm^2 .

OR

Contd...4

